

Programme	B. Sc. Computer Science				
Course Title	Deep Learning				
Type of Course	Elective				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	4. Introduction to Artificial Intelligence 5. Basic understanding of linear algebra, calculus, and probability. 6. Basics of Machine learning				
Course Summary	The theoretical groundwork for comprehending the fundamentals of deep learning is supplied by this course. Theoretical frameworks, optimisation techniques, and mathematical ideas that support deep neural network building and training will be examined by students.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Master key concepts of machine learning, understanding various layers of neural network.	U	C	Instructor-created exams / Quiz
CO2	Understand and implement the backpropagation algorithm for training neural networks, demonstrating the ability to compute gradients and update weights.	Ap, U	P	Assignment / Seminar presentations/ Exams
CO3	Analyze and compare different activation functions used in neural networks, explaining their role in the learning process.	U	P	Seminar Presentation / Group Tutorial Work/ Viva Voce
CO4	Design and implement feedforward neural networks for various applications, considering aspects such as model architecture, activation functions, and initialization methods.	Ap	C	Instructor-created exams / Home Assignments
CO5	Master the principles of convolutional neural networks, including convolutional layers, pooling layers, and their applications in computer vision. Master various regularization techniques, such as dropout, batch normalization, and weight regularization, to improve the generalization of neural networks	U	P	Writing assignments/ Exams/ Seminar Presentations

CO6	Apply deep learning concepts to solve real-world problems, demonstrating the ability to choose appropriate architectures and hyperparameters.	Ap	P	Case Study
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Machine Learning Basics		10	15
	1	Learning Algorithms -Supervised learning- regression, classification, Unsupervised learning, Reinforcement learning (Introduction only)	2	
	2	Terms - Capacity, Overfitting and Underfitting, Hyperparameters and Validation Sets, Estimators, Bias and Variance	2	
	3	Maximum Likelihood estimation, Bayesian statistics, Stochastic Gradient Descent	3	
	4	Building a Machine Learning Algorithm	1	
	5	Challenges Motivating Deep Learning	2	
II	Optimisation and Neural Networks		15	20
	6	Neural Networks –Perceptron, Gradient Descent solution for Perceptron, Multilayer perceptron	3	
	7	Activation Functions- Sigmoid, Softmax, Relu, LeakyRelu, ERELU	2	
	8	Chain rule, back propagation- Backpropagation Algorithm	3	
	9	Gradient based learning.	2	
	10	Introduction to optimization– Gradient based optimization, linear least squares. Stochastic gradient descent	2	
	11	Regularisation techniques- Drop out, Batch Normalisation, weight regularisation	3	
III	Convolutional Neural Network		12	20
	12	Convolutional Neural Networks – convolution operation, motivation	2	
	13	Pooling	2	
	14	Variants of convolution functions	2	
	15	Structured outputs, data types	2	
	16	CNN Architecture- Alexnet, VGG16	4	
IV	Deep learning Architectures		11	15
	17	Sequence Modeling: Recurrent and Recursive Nets- Basics of Recurrent Neural Networks	2	
	18	Encoder – Decoder Sequence to Sequence Architectures,	2	
	19	Deep Recurrent Networks, Recursive Neural Networks	2	
	20	The Long Short-Term Memory	2	
	21	GRU	2	

	22	Basics of transfer learning techniques (Concept only)	1	
V	Open ended Module		12	
	1	- Master students Basics of Mathematics required for Machine learning and deep learning- Linear Algebra (Scalars, Vectors, Matrices and Tensors, Eigen values, Eigen Vectors)- concepts only - Probability awareness- Why probability, random variable, probability distributions)- concepts only - Discuss advanced topics in deep learning, including transfer learning, autoencoders, adversarial training, and stay informed about recent developments in the field.)- concepts only		

References

- "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville.
- Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, "Dive into Deep Learning", August 2019.
- Neural Networks and Deep Learning: A Textbook by Charu C. Aggarwal. Springer. 1st edition, 2018.
- "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	-	-	-	2	3						
CO 2	2	-	1	1	2	3						
CO 3	2	-	-	-	2	1						
CO 4	2	-	1	1	2	2						
CO 5	2	-	2	1	2	3						
CO 6	2	-	2	1	2	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations
CO 1	✓		✓
CO 2	✓		✓
CO 3	✓		✓
CO 4		✓	✓
CO 5		✓	✓
CO 6		✓	